Why people punish – Simulation

1. Setup

Individuals are endowed with a strategy for the repeated prisoner's dilemma, represented by a finite-state automaton (see Section 2). They are assigned to groups of size $N = 6$ and are randomly rematched for $T = 20$ supergames.

At the start of each match, players begin in state 1 and play an indefinitely repeated prisoner's dilemma with a punishment option. The continuation probability of the repeated game is $\delta = 0.7$.

The stage-game matrix is parametrized by:

- b and c: benefits and costs from cooperation;
- α and β: cost and damage of punishment.

The resulting payoff matrix is is:

	C	D	P
C	b-c	-c	-c- β
D	b	0	- β
P	b-α	-α	-α- β

To avoid negative payoffs, I apply a linear transformation with parameters l and m:

	C	D	P
C	l(b-c)+m	l(-c)+m	l(-c- β)+m
D	l(b)+m	m	l(- β)+m
P	l(b-α)+m	l(-α)	l(-α- β)+m

We consider two treatments:

	Non-cooperative treatment	Cooperative treatment
b	4	2
c	3	1
α	3	1
β	12	4
l	1	3

m	15	15

So that the resulting payoff matrices are, for the cooperative treatment:

	C	D	P
C	18	12	0
D	21	15	3
P	18	12	0

And for the non-cooperative treatment:

	C	D	P
C	16	12	0
D	19	15	3
P	16	12	0

2. Strategies

Each subject is characterized by a strategy. Strategies determine behavior within a supergame.

We focus on three types:

1. non-punishing cooperator (grim-trigger);
2. punishing cooperator;
3. defector.

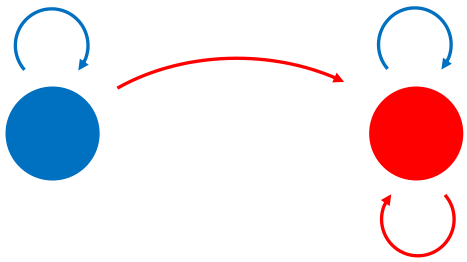
The initial distribution of strategies in the group varies across sessions and is given by: $(N_c,\ N_P,\ N_D)$, representing respectively the number of non-punishing cooperators, punishing cooperators and defectors.

Strategies are represented by finite-state automata. Each auotomaton is represented (like in Chris' simulations) like a matrix where:

- Each column represent a state;
- The first row encodes the action in each state;
- Each of the following rows encodes the destination state after a certain pair of actions has been observed.

	STATE 1	STATE 2	STATE 3	...
ACTION	Action in state 1	Action in state 2	Action in state 3	
CC	If in state 1 and CC observed, which state to transition to	If in state 2 and CC observed, which state to transition to	...	
CD	If in state 1 and CD observed, which state to transition to	...		
CP	...			
DC				
DD				
DP				
PC				
PD				
PP				

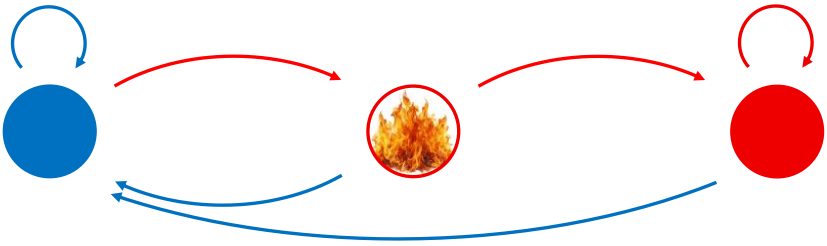
2.1 Non-punishing cooperator (Grim-Trigger)



	STATE 1	STATE 2
ACTION	C	D
CC	1	
CD	2	
CP	2	
DC		2
DD		2

DP		2
PC		
PD		
PP		

2.2 Punishing cooperators



	STATE 1	STATE 2	STATE 3
ACTION	C	P	D
CC	1		
CD	2		
CP	2		
DC			1
DD			3
DP			3
PC		1	
PD		3	
PP		3	

2.3 Defector

	STATE 1
ACTION	D
CC	
CD	

CP	
DC	1
DD	1
DP	1
PC	
PD	
PP	

3. Meta-strategies

At the end of each supergame, one randomly selected individual in the group may revise their strategy.

The update rule is as follows:

- The individual evaluates all supergames since their last update
- They compute their average per-round payoff
- They compare it with counterfactual payoffs from the two alternative strategies[1]
- If one alternative yields a strictly higher payoff, they switch to that strategy
- If both alternatives yield equal payoffs, the choice is random

Additionally, punishers receive a payoff adjustment; their average per-round payoff is increased by a term "punisher_bias" (default set at 1).

4. Data generation

I generate data for 12 sessions. In each session there are 4 matching groups. Each group is randomly composed, and in particular:

$$N_c \sim U[1,3]$$
$$N_N \sim U[1,3]$$
$$N_D = N - N_C - N_N$$

Within each session, two groups are assigned to the cooperative treatment, and two to the non-cooperative treatment.

The resulting simulated dataset has the following columns:

Column	Explanation
session_id	Identifier for the session
group_id	Identifier for the group
treatment	Either "cooperative" or "non-cooperative", with two groups per treatment per session

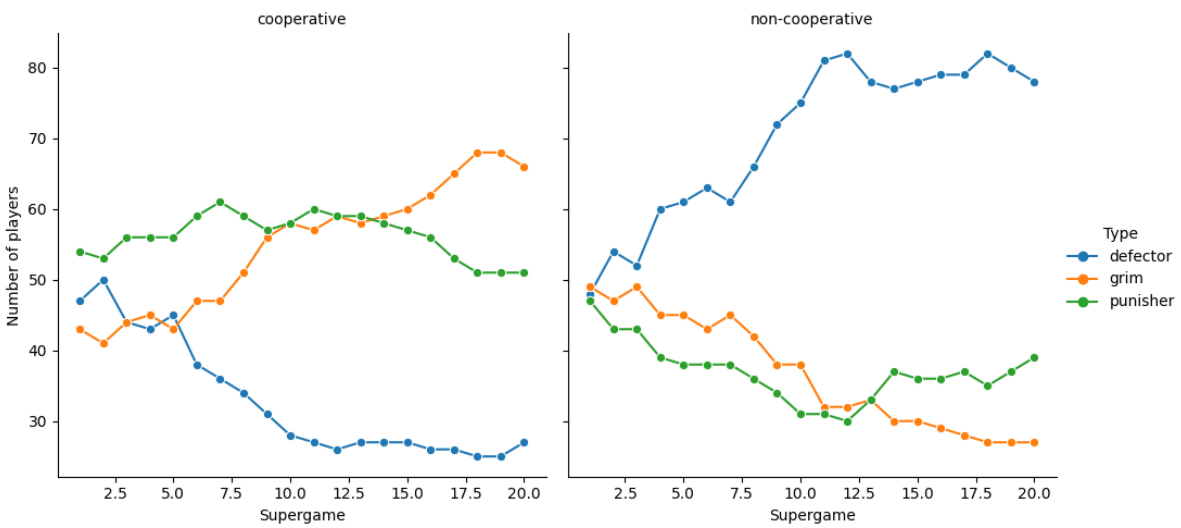
[1] The counterfactual payoff is computed by holding the length of each supergame equal to its realized value, while allowing action profiles to differ. We assume that the agent knows the types of her opponents and recomputes the payoff she would have obtained under an alternative strategy, given opponents' strategies and the realized duration of each round.

Nc	Number of non-punishing cooperators at the beginning of a session
Np	Number of punishing cooperators at the beginning of a session
Nd	Number of defectors at the beginning of a session
supergame	Number of the supergame
pair_id	Identifier of the current pair
round	Current round in the supergame
player_id	Identifier of the player in the group
partner_id	Identifier of the opponent in the group
type_start	Player type at the beginning of the session
type_current	Current player type
action	Action played in that round (either C,D or P)
opponent_action	Opponent's action in that round
payoff	Payoff for that round
cum_payoff	Cumulative payoff

5. Analysis (data_new.csv)

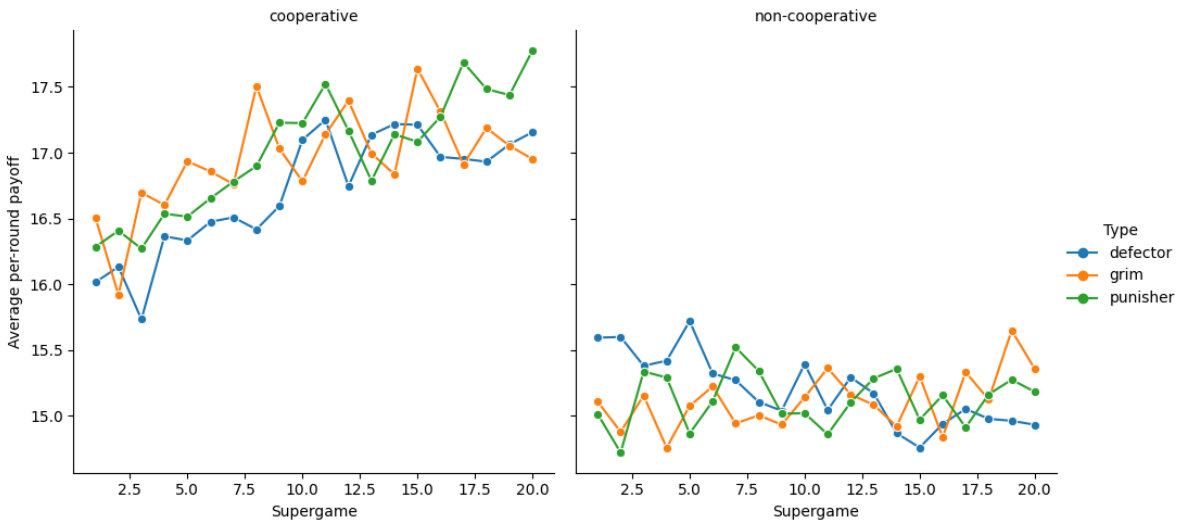
5.1 Strategies count over supergames

The following figure plots a count of how many individuals adopt each of the three strategies in the two treatments over the supergames.



5.2 Average per-round payoff over supergames

The following figure plots the average payoff for each type in the two treatments over the supergames. The average is obtained by: (i) computing the individual per-round average payoff in each supergame, (ii) averaging over individuals with the same starting type.



5.3 Regression results

For each treatment, I first discard observations belonging to defectors.

5.3.1 Simple regression

I then run the following regression:

$$y_{is} = \beta_0 + \beta_1 P_i + \varepsilon_{is}$$

where y_{is} is individual's i average per-round payoff in supergame s and P_i is a dummy indicating whether i is a punisher at the beginning of the game. SE are clustered at the individual level.

Results for the **cooperative treatment**:

```
OLS Regression Results
==============================================================================
Dep. Variable:          average_payoff   R-squared:                       0.000
Model:                             OLS   Adj. R-squared:                 -0.000
Method:                  Least Squares   F-statistic:                    0.1129
Date:                 Wed, 08 Apr 2026   Prob (F-statistic):              0.738
Time:                         11:53:40   Log-Likelihood:                -4226.1
No. Observations:                 1940   AIC:                             8456.
Df Residuals:                     1938   BIC:                             8467.
Df Model:                            1
Covariance Type:               cluster
==============================================================================
coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     16.9508      0.124    137.062      0.000      16.708      17.193
P              0.0571      0.170      0.336      0.737      -0.276       0.390
==============================================================================
Omnibus:                      308.765   Durbin-Watson:                   1.703
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              469.332
Skew:                          -1.178   Prob(JB):                    1.22e-102
Kurtosis:                       3.509   Cond. No.                         2.77
==============================================================================

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)
```

Results for the **non-cooperative treatment**:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          average_payoff   R-squared:                       0.000
Model:                             OLS   Adj. R-squared:                 -0.001
Method:                  Least Squares   F-statistic:                  0.003507
Date:                 Wed, 08 Apr 2026   Prob (F-statistic):              0.953
Time:                         11:53:40   Log-Likelihood:                -3667.8
No. Observations:                 1920   AIC:                             7340.
Df Residuals:                     1918   BIC:                             7351.
Df Model:                            1
Covariance Type:               cluster
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     15.1170      0.083    182.660      0.000      14.955      15.279
P              0.0078      0.132      0.059      0.953      -0.252       0.267
==============================================================================
Omnibus:                       48.529   Durbin-Watson:                   2.061
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               59.226
Skew:                          -0.318   Prob(JB):                     1.38e-13
Kurtosis:                       3.579   Cond. No.                         2.59
==============================================================================

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)
```

5.3.2 Controlling for group composition – assuming linear effects

I then run the following regression:

$$y_{is} = \beta_0 + \beta_1 P_i + \beta_2 N_d + \beta_3 N_p + \beta_4 N_d * P_i + \beta_5 N_P * P_i + \varepsilon_{is}$$

Np and Nd represent respectively how many other punishers and defectors are in the group. Both are considered here as numeric variables (linear effects assumed).

Results for the **cooperative treatment**:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          average_payoff   R-squared:                       0.043
Model:                             OLS   Adj. R-squared:                  0.041
Method:                  Least Squares   F-statistic:                     8.597
Date:                 Wed, 08 Apr 2026   Prob (F-statistic):           9.26e-07
Time:                         14:50:57   Log-Likelihood:                -4183.2
No. Observations:                 1940   AIC:                             8378.
Df Residuals:                     1934   BIC:                             8412.
Df Model:                            5
Covariance Type:               cluster
==================================================================================
                   coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept       15.8935      0.597     26.624      0.000      14.724      17.064
P                0.6350      0.766      0.829      0.407      -0.867       2.137
Np_others        0.4893      0.178      2.746      0.006       0.140       0.838
Nd_others       -0.0550      0.120     -0.459      0.646      -0.290       0.180
P:Np_others      0.0228      0.260      0.088      0.930      -0.487       0.532
P:Nd_others     -0.1126      0.172     -0.655      0.512      -0.449       0.224
==============================================================================
Omnibus:                      290.010   Durbin-Watson:                   1.775
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              429.607
```

```
Skew:                          -1.105   Prob(JB):                    5.15e-94
Kurtosis:                       3.656   Cond. No.                        46.2
==============================================================================

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)
```

Results for the **non-cooperative treatment**:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:         average_payoff   R-squared:                       0.045
Model:                            OLS   Adj. R-squared:                  0.042
Method:                 Least Squares   F-statistic:                     13.02
Date:                Wed, 08 Apr 2026   Prob (F-statistic):           1.18e-09
Time:                        14:50:57   Log-Likelihood:                -3624.1
No. Observations:                1920   AIC:                             7260.
Df Residuals:                    1914   BIC:                             7294.
Df Model:                           5
Covariance Type:              cluster
==================================================================================
                   coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept       15.7882      0.422     37.423      0.000      14.961      16.615
P               -0.5037      0.550     -0.916      0.360      -1.582       0.574
Np_others       -0.0580      0.135     -0.430      0.667      -0.322       0.206
Nd_others       -0.3259      0.092     -3.560      0.000      -0.505      -0.146
P:Np_others      0.2033      0.203      1.001      0.317      -0.195       0.601
P:Nd_others      0.0950      0.133      0.716      0.474      -0.165       0.355
==============================================================================
Omnibus:                       60.882   Durbin-Watson:                   2.148
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               97.202
Skew:                          -0.284   Prob(JB):                     7.81e-22
Kurtosis:                       3.945   Cond. No.                        36.4
==============================================================================

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)
```

5.3.3 Controlling for group composition – not assuming linear effects

In order not to assume linear effects without having to control for each single group composition, I first create two coarse categorical variables:

1. "Np_categorical", which assumes value "fewP" if the group contains 0 or 1 (*additional*) punishers and value "manyP" otherwise;
2. "Nd_categorical", which assumes value "fewD" if the group contains 0, 1 or 2 defectors and value "manyD" otherwise.

I then run the regression:

$$y_{is} = \beta_0 + \beta_1 P_i + \beta_2 N_{d,categorical} + \beta_3 N_{p,categorical} + \beta_4 N_{d,categorical} * P_i + \beta_5 N_{P,categorical} * P_i + \varepsilon_{is}$$

still with SE clustered at the individual level.

Results for the **cooperative treatment**:

```
OLS Regression Results
==============================================================================
Dep. Variable:         average_payoff   R-squared:                       0.039
Model:                            OLS   Adj. R-squared:                  0.037
Method:                 Least Squares   F-statistic:                     6.616
Date:                Wed, 08 Apr 2026   Prob (F-statistic):           2.48e-05
Time:                        17:06:14   Log-Likelihood:                -4187.4
No. Observations:                1940   AIC:                             8387.
Df Residuals:                    1934   BIC:                             8420.
Df Model:                           5
Covariance Type:              cluster
==========================================================================================
                                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
Intercept                     16.3672      0.247     66.354      0.000      15.884      16.851
C(Np_categorical)[T.manyP]     0.7538      0.265      2.843      0.004       0.234       1.273
C(Nd_categorical)[T.manyD]    -0.3495      0.263     -1.331      0.183      -0.864       0.165
P                              0.4994      0.324      1.542      0.123      -0.136       1.134
P:C(Np_categorical)[T.manyP]  -0.2242      0.356     -0.630      0.529      -0.922       0.473
P:C(Nd_categorical)[T.manyD]  -0.2858      0.419     -0.682      0.495      -1.107       0.535
==============================================================================
Omnibus:                      291.126   Durbin-Watson:                   1.772
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              431.824
Skew:                          -1.109   Prob(JB):                     1.70e-94
Kurtosis:                       3.648   Cond. No.                         14.6
==============================================================================

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)
```

Results for the **non-cooperative treatment**:

```
OLS Regression Results
==============================================================================
Dep. Variable:         average_payoff   R-squared:                       0.024
Model:                            OLS   Adj. R-squared:                  0.021
Method:                 Least Squares   F-statistic:                     4.000
Date:                Wed, 08 Apr 2026   Prob (F-statistic):            0.00245
Time:                        17:06:14   Log-Likelihood:                -3644.5
No. Observations:                1920   AIC:                             7301.
Df Residuals:                    1914   BIC:                             7334.
Df Model:                           5
Covariance Type:              cluster
==========================================================================================
                                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
Intercept                     14.9637      0.173     86.738      0.000      14.626      15.302
C(Np_categorical)[T.manyP]     0.3600      0.201      1.793      0.073      -0.034       0.753
C(Nd_categorical)[T.manyD]    -0.1539      0.212     -0.725      0.469      -0.570       0.262
P                             -0.0600      0.319     -0.188      0.851      -0.685       0.565
P:C(Np_categorical)[T.manyP]   0.0655      0.354      0.185      0.853      -0.629       0.760
P:C(Nd_categorical)[T.manyD]  -0.1094      0.399     -0.274      0.784      -0.891       0.672
==============================================================================
Omnibus:                       59.178   Durbin-Watson:                   2.117
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               85.568
Skew:                          -0.309   Prob(JB):                     2.63e-19
Kurtosis:                       3.830   Cond. No.                         12.1
==============================================================================

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)
```